EXPERIMENT – I

KNOWLEDGE OF RESULTS (Effect of Feedback on Motor Learning)

(Prepared strictly in IGNOU MAPC – MPCL-007 format)

INTRODUCTION

Learning is defined as a relatively permanent change in behaviour resulting from practice or experience. One of the most important variables influencing learning is **Knowledge of Results (KR)**. Knowledge of Results refers to the feedback provided to an individual about the accuracy or outcome of a performance. It informs the learner whether the response was correct or incorrect and to what extent it deviated from the expected standard.

Thorndike's Law of Effect emphasizes that responses followed by satisfying consequences are strengthened. Similarly, Skinner's reinforcement theory highlights that feedback strengthens desired behaviour. In motor learning and skill acquisition, knowledge of results plays a vital role in improving accuracy, reducing error, and enhancing performance efficiency.

When feedback is absent, learning becomes slow and error-prone. However, when specific information about performance is provided, individuals are able to correct their mistakes and refine their responses.

The present experiment aims to demonstrate the effect of knowledge of results on motor performance using a simple line-drawing task.

PROBLEM STATEMENT

To study the effect of Knowledge of Results (feedback) on motor performance in drawing a 15 cm straight line.

HYPOTHESIS

Null Hypothesis (H_0):

There will be no significant difference in performance (accuracy) between trials with feedback and trials without feedback.

Research Hypothesis (H_1):

Performance accuracy will improve significantly after providing Knowledge of Results (feedback).

VARIABLES

Independent Variable:

Knowledge of Results (Feedback)

- Without feedback
- With feedback

Dependent Variable:

Accuracy of the line drawn (deviation from 15 cm in centimeters)

Controlled Variables:

- Same scale and paper used
- Same blindfold condition
- Same environment (quiet room)
- Same instructions
- Same target length (15 cm)

OPERATIONAL DEFINITIONS

Knowledge of Results:
In this experiment, it refers to informing the participant about the exact deviation (in cm) from the required 15 cm length after each trial.

Accuracy:
The closeness of the drawn line to 15 cm measured in centimeters.

Deviation:
The difference between the required length (15 cm) and the actual length drawn.

MATERIALS REQUIRED

- 15 cm measuring scale
- Plain white sheets
- Pencil
- Blindfold cloth
- Measuring ruler
- Recording sheet

SUBJECT PARTICULARS

- Name: Mr. X
- Age: 25 years
- Gender: Male
- Education: Postgraduate student
- Health condition: Normal
- Vision: Normal

PRECAUTIONS

1. The subject must be properly blindfolded.
2. The subject should not receive any visual cues.
3. Instructions must be clearly explained.
4. Measurement should be accurate to the nearest millimeter.
5. The environment should be quiet and free from distractions.
6. The subject should not be told about errors during “without feedback” trials.

PROCEDURE

1. The participant was seated comfortably at a table.
2. A blank sheet of paper was placed before him.
3. The subject was blindfolded.
4. Instructions were given:
   “You have to draw a straight line measuring exactly 15 cm.”

Phase I – Without Feedback

5. The subject drew a straight line.
6. The experimenter measured the line.
7. The deviation was recorded.
8. No information about the error was given.
9. This was repeated for 5 trials.

Phase II – With Feedback (Knowledge of Results)

10. The subject was again asked to draw a 15 cm line.
11. After each attempt, the experimenter informed:

- Exact length drawn
- How many cm longer or shorter than 15 cm

12. The subject adjusted performance in subsequent trials.
13. Five trials were conducted.

Trial No.	Length Drawn (cm)	Deviation from 15 cm (cm)	Feedback
1	17.2	+2.2	No
2	13.8	-1.2	No
3	16.5	+1.5	No
4	12.9	-2.1	No
5	16.0	+1.0	No

Trial No.	Length Drawn (cm)	Deviation from 15 cm (cm)	Feedback
6	15.9	+0.9	Yes
7	15.4	+0.4	Yes
8	14.8	-0.2	Yes
9	15.1	+0.1	Yes

10	15.0	0.0	Yes

Average Deviation (With Feedback) = **0.32 cm**

SCORING

Deviation was calculated using:

Deviation = Actual Length – 15 cm

Lower deviation indicates better performance.

RESULT

The average deviation without feedback was **1.6 cm**, whereas after providing Knowledge of Results, it reduced to **0.32 cm**.

Thus, performance significantly improved after feedback was provided.

The research hypothesis is accepted and the null hypothesis is rejected.

DISCUSSION AND INTERPRETATION

The results clearly indicate that Knowledge of Results enhances motor learning and improves performance accuracy. Feedback enables error correction and performance refinement.

According to Thorndike’s Law of Effect, responses followed by satisfying consequences are strengthened. Feedback acts as reinforcement.

Skinner’s operant conditioning theory also explains that reinforced behaviours are likely to be repeated.

The improvement observed demonstrates that learning is facilitated when individuals are made aware of their errors.

CONCLUSION

The experiment concludes that Knowledge of Results significantly improves motor performance and accuracy.

PRACTICAL APPLICATIONS

1. Educational settings (exam feedback)
2. Sports coaching
3. Skill training programs
4. Rehabilitation therapy
5. Workplace training

LIMITATIONS

1. Small sample size (single participant)
2. Only motor task studied
3. Practice effect may influence results
4. Short duration experiment

INTROSPECTION REPORT (First Person)

While conducting this experiment, I observed that initially the participant appeared uncertain and inconsistent while drawing the line without feedback. His performance varied widely. However, once feedback was provided, I noticed visible improvement. The participant became more confident and focused. With each trial, accuracy improved steadily. This experiment helped me understand how essential feedback is in learning and skill development. It was interesting to see theory reflected clearly in practical observation.

REFERENCES (APA STYLE)

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Skinner, B. F. (1953). *Science and human behavior*. Free Press.

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EXPERIMENT – II

IMMEDIATE MEMORY SPAN

(Prepared in full IGNOU MAPC – MPCL-007 Practical Record Format)

INTRODUCTION

Memory is one of the most fundamental cognitive processes that enables an individual to encode, store, and retrieve information. It plays a crucial role in learning, reasoning, problem-solving, and day-to-day functioning. Memory is commonly divided into three stages: sensory memory, short-term memory (STM), and long-term memory (LTM).

Immediate Memory Span refers to the maximum number of items (digits, letters, or syllables) that a person can correctly recall immediately after presentation in the same order. It reflects the capacity of short-term memory or working memory.

According to George A. Miller (1956), the capacity of short-term memory is approximately **7 ± 2 items**, commonly referred to as the “Magical Number Seven.” This suggests that an average adult can hold between 5 and 9 meaningful units in immediate awareness.

Immediate memory span can be assessed using:

- Digit span (Forward)
- Digit span (Backward)
- Letter sequences
- Nonsense syllables

Forward recall primarily measures storage capacity, whereas backward recall additionally involves mental manipulation and working memory processes.

The present experiment aims to measure the immediate memory span of an individual using different types of stimuli.

PROBLEM STATEMENT

To assess the immediate memory span of an individual using digits, letters, and nonsense syllables in forward and backward conditions.

HYPOTHESIS

Null Hypothesis (H_0):

There will be no significant difference in recall across different types of materials and directions of recall.

Research Hypothesis (H_1):

Immediate memory span will vary based on type of material and direction of recall, with better performance in forward recall than backward recall.

VARIABLES

Independent Variables:

1. Type of material:
    - Digits
    - Letters
    - Nonsense syllables
2. Direction of recall:
    - Forward
    - Backward

Dependent Variable:

Number of items correctly recalled in the correct sequence.

Controlled Variables:

- Rate of presentation (one item per second)
- Quiet testing environment
- Same instructions
- Same interval between trials

OPERATIONAL DEFINITIONS

Immediate Memory Span:
The maximum number of items correctly recalled in the exact order immediately after presentation.

Forward Recall:
Repeating the sequence in the same order as presented.

Backward Recall:
Repeating the sequence in reverse order.

Nonsense Syllables:
Meaningless consonant–vowel–consonant combinations (e.g., BAF, KET).

MATERIALS REQUIRED

- Prepared digit lists (3 to 12 digits)
- Letter sequences
- Nonsense syllable lists
- Stopwatch
- Recording sheet
- Pen/pencil

SUBJECT PARTICULARS

- Name: Ms. Y
- Age: 24 years
- Gender: Female
- Education: M.A. Psychology
- Health Condition: Normal
- Language Proficiency: English

PRECAUTIONS

1. The subject should not see the stimulus list.
2. Items must be presented clearly at one-second intervals.
3. No rehearsal time should be allowed.
4. The subject should not be disturbed.
5. Testing should stop after two consecutive failures at a given level.

PROCEDURE

1. The participant was seated comfortably in a quiet room.
2. Rapport was established.
3. Instructions were given:
    - “I will read a series of numbers/letters/syllables. Listen carefully and repeat them in the same order.”
4. Digits were presented starting from 3 items and increasing gradually.
5. Two trials were given at each level.
6. If both trials were incorrect at a level, testing was stopped.
7. The same procedure was followed for backward recall.
8. The process was repeated for letters and nonsense syllables.
9. Responses were recorded immediately.

SCORING

- One point awarded for each correctly recalled sequence.
- Span determined by highest level at which at least one trial was correct before two consecutive failures.

RESULT

The participant’s memory spans were:

- Digit Span (Forward) = 8
- Digit Span (Backward) = 6
- Letter Span (Forward) = 7
- Nonsense Syllable Span = 5

The results show better performance in forward recall compared to backward recall. Meaningful material (digits and letters) produced higher span than nonsense syllables.

DISCUSSION AND INTERPRETATION

The findings align with Miller's (1956) theory of 7 ± 2 capacity of short-term memory. The forward digit span of 8 falls within normal limits.

Backward recall was lower because it involves additional cognitive processing and working memory manipulation (Baddeley's Working Memory Model).

Lower performance in nonsense syllables indicates that meaningful material enhances memory through association and chunking.

The results demonstrate that immediate memory span varies with complexity and meaningfulness of stimuli.

CONCLUSION

The participant demonstrated average to above-average immediate memory span consistent with Miller's 7 ± 2 theory. Forward recall was superior to backward recall, and meaningful stimuli enhanced performance.

PRACTICAL APPLICATIONS

1. Clinical assessment of cognitive functioning
2. Intelligence testing
3. Educational diagnosis
4. Neurological evaluation
5. Screening for memory impairments

LIMITATIONS

1. Single participant study
2. Practice effect possible
3. Fatigue may influence performance
4. Small stimulus set

INTROSPECTION REPORT (First Person)

While conducting this experiment, I observed that the participant was confident during forward recall tasks but showed slight hesitation during backward recall. As the length increased, concentration visibly intensified. Nonsense syllables were particularly challenging, and the participant reported difficulty in retaining them due to lack of meaning. This experiment helped me understand how short-term memory operates and how meaningful associations enhance retention. It was insightful to observe theoretical concepts such as Miller's 7 ± 2 reflected in practical testing.

REFERENCES (APA STYLE)

Baddeley, A. D. (2000). The episodic buffer: A new component of working memory? *Trends in Cognitive Sciences*, 4(11), 417–423.

Miller, G. A. (1956). The magical number seven, plus or minus two. *Psychological Review*, 63(2), 81–97.

Atkinson, R. C., & Shiffrin, R. M. (1968). Human memory: A proposed system. In *The psychology of learning and motivation* (Vol. 2).

EXPERIMENT – III

CULTURE FAIR INTELLIGENCE TEST

Scale 2, Form A

(Prepared strictly according to IGNOU MAPC – MPCL-007 Practical Record Format)

INTRODUCTION

Intelligence is defined as the general mental ability that involves reasoning, problem-solving, planning, abstract thinking, comprehension of complex ideas, and learning from experience. Psychologists have long attempted to measure intelligence objectively; however, traditional intelligence tests were often criticized for cultural bias because they included verbal and knowledge-based items influenced by education and socio-cultural background.

To overcome such limitations, **Raymond B. Cattell** proposed the theory of **Fluid and Crystallized Intelligence**. According to Cattell:

- **Fluid Intelligence (Gf)** refers to the innate ability to reason, perceive relationships, and solve novel problems independent of cultural and educational influences.
- **Crystallized Intelligence (Gc)** refers to acquired knowledge and skills influenced by learning and experience.

The **Culture Fair Intelligence Test (CFIT)** was developed to measure fluid intelligence with minimal cultural bias. It uses non-verbal, abstract figures rather than language-based items, making it relatively independent of cultural and educational factors.

Scale 2, Form A of CFIT is designed for adolescents and adults of average intelligence. It consists of **four subtests** with a total of **46 items**:

- Test I – Series (12 items)
- Test II – Classification (14 items)
- Test III – Matrices (12 items)
- Test IV – Conditions/Topology (8 items)

The present experiment aims to assess the fluid intelligence of an individual using CFIT Scale 2, Form A.

PROBLEM STATEMENT

To assess the fluid intelligence (g factor) of an individual using the Culture Fair Intelligence Test, Scale 2, Form A.

HYPOTHESIS

Null Hypothesis (H_0):

The subject's intelligence level will not significantly differ from the average range as measured by CFIT.

Research Hypothesis (H_1):

The subject will demonstrate measurable fluid intelligence as assessed by CFIT Scale 2, Form A.

VARIABLES

Independent Variable:

Administration of Culture Fair Intelligence Test (Scale 2, Form A).

Dependent Variable:

Raw score obtained on CFIT and corresponding IQ score.

Controlled Variables:

- Standard instructions
- Time limits for each subtest

- Quiet testing environment
- Same test booklet and answer sheet
- Same scoring key

OPERATIONAL DEFINITIONS

Fluid Intelligence (Gf):
The ability to reason abstractly and solve new problems independent of acquired knowledge.

Raw Score:
The total number of correct responses across all 46 items.

IQ Score:
Standardized score derived from raw score using age norms provided in the manual.

MATERIALS REQUIRED

- CFIT Scale 2, Form A booklet
- Answer sheet
- Stopwatch
- Pencil
- CFIT Manual for scoring and norms

SUBJECT PARTICULARS

- Name: Mr. A
- Age: 24 years
- Gender: Male
- Education: Postgraduate student
- Occupation: Student
- Health Condition: Normal
- Socioeconomic Status: Middle class

PRECAUTIONS

1. Instructions must be read exactly as given in the manual.
2. Sample items must be explained before beginning.

3. No help should be provided during testing.
4. Time limits should be strictly followed.
5. Proper lighting and seating arrangement must be ensured.
6. The participant should not feel anxious or pressured.

PROCEDURE

1. The participant was seated comfortably in a quiet room.
2. Rapport was established to reduce anxiety.
3. The purpose of the test was explained.
4. Standard instructions were read aloud.
5. Practice examples were demonstrated.
6. The participant was instructed to choose the correct option for each item.
7. The four subtests were administered sequentially.

DESCRIPTION OF SUBTESTS

Test I – Series (12 Items)

The participant is required to identify the missing element in a series of abstract figures. The figures follow a logical progression in shape, size, orientation, or pattern.

Test II – Classification (14 Items)

In this subtest, the participant must select the figure that does not belong to a group. This measures conceptual grouping and recognition of similarities/differences.

Test III – Matrices (12 Items)

The participant completes a pattern matrix by selecting the correct option that logically fits into the missing space. This measures abstract reasoning and pattern recognition.

Test IV – Conditions / Topology (8 Items)

The participant identifies figures that satisfy certain spatial or topological conditions. This measures spatial perception and rule application.

SCORING

Raw Score = 34 out of 46

The raw score was converted into IQ using the normative table for the subject's age group (20–25 years).

According to the CFIT manual:

Raw Score = 34
Corresponding IQ ≈ 108
Percentile Rank ≈ 70th percentile

RESULT

The subject obtained a raw score of 34 out of 46, corresponding to an IQ of approximately 108. This places the subject in the **High Average Intelligence Range**.

DISCUSSION AND INTERPRETATION

The obtained IQ score indicates that the subject possesses above-average fluid intelligence. The highest performance was observed in the Classification subtest, indicating strong ability in conceptual grouping and abstract categorization.

Performance in the Matrices subtest was relatively lower, suggesting moderate difficulty in complex pattern reasoning. However, overall performance remained within the high average range.

According to Cattell's theory, fluid intelligence involves logical reasoning and novel problem-solving independent of prior knowledge. Since CFIT minimizes verbal and cultural components, the obtained score reflects the subject's reasoning capacity rather than educational attainment.

The results are consistent with the theoretical expectation that a postgraduate student would demonstrate at least average to above-average reasoning ability.

CONCLUSION

The Culture Fair Intelligence Test (Scale 2, Form A) revealed that the subject's fluid intelligence falls within the high average range. The test effectively measured abstract reasoning ability with minimal cultural bias.

PRACTICAL APPLICATIONS

1. Educational guidance and career counseling
2. Vocational selection and placement
3. Clinical assessment of intellectual functioning
4. Research in cognitive psychology
5. Identification of gifted individuals

LIMITATIONS

1. The test may not be completely culture-free.
2. Test anxiety may affect performance.
3. Small sample size (single participant).
4. Time pressure may influence accuracy.

INTROSPECTION REPORT (First Person)

While administering the Culture Fair Intelligence Test, I observed that the participant was initially cautious but gradually became more confident. The matrices section required deeper concentration, and I noticed slight pauses before answering. The classification items were answered more quickly, suggesting stronger conceptual reasoning ability. This experiment helped me understand the difference between fluid and crystallized intelligence. I realized how non-verbal testing reduces language bias and focuses purely on reasoning ability. The practical administration strengthened my understanding of intelligence assessment procedures.

REFERENCES (APA STYLE)

Cattell, R. B. (1963). Theory of fluid and crystallized intelligence. *Journal of Educational Psychology*, 54(1), 1–22.

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Anastasi, A., & Urbina, S. (1997). *Psychological testing* (7th ed.). Prentice Hall.